

Advanced FIB sample preparation techniques for high resolution TEM investigations of HEMT structures



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ABSTRACT

In this paper advanced sample preparation techniques based on focused ion beam (FIB) optimized for TEM investigation of high electron mobility transistor (HEMT) structures are presented. It is shown that the usage of an innovative in-situ lift-out method combined with X2 window and backside milling techniques as well as live thickness control and end point detection can significantly improve the quality of electron transparent samples required for high resolution TEM investigations. This advanced preparation flow is evaluated and demonstrated at GaN HEMT structures for atomic resolution TEM investigation.

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1. Introduction

III/V Semiconductor based high electron mobility transistor (HEMT) devices provide an outstanding performance for high frequency and high power applications used for radar and telecommunication systems. The devices are based on III/V heterostructures to realize high electron mobility in the 2DEG. The electron channel is contacted by ohmic source and drain metal structures. The channel switching is realized by a sub μm sized metal gate structure forming a Schottky contact.

The electrical functionality, performance and reliability of HEMT devices is strongly related to crystal defects and interface perfection within the epitaxial grown heterostructure in particular the 2DEG channel, the geometry and interface quality of the metal gate as well as the metal diffusion into the substrate at the ohmic contacts. Therefore high resolution transmission electron microscopy (TEM) is the method of choice for device characterization required for technology qualification and failure root cause analysis.

Typically focused ion beam (FIB) sample preparation is applied to prepare electron transparent samples from the small sized transistor structures and related defect sites. But FIB lamella preparation is challenging because of the heterogeneous device structure including many different materials and inhomogeneous topography producing curtaining and following thickness variation during

FIB milling. On the other side high quality almost artefact free TEM lamellas with homogeneous thickness below 50 nm are required for high resolution TEM.

An advanced FIB sample preparation flow is developed implementing new in-situ lift-out, X2 window milling [1], backside milling techniques and end point detection of the lamella thickness [2] to achieve homogeneous ultra-thin and almost artefact free TEM lamellas.

2. In-situ lift-out by micro grippers and clamp holders

Nowadays in situ lift out by using W needles and FIB deposition is standard technique to transfer TEM lamellas out of the sample. After pre-thinning of a lamella a W needle is attached to the lamella and welded by in-situ beam induced deposition. Afterwards the lamella is cut from the sample and transferred to a sample holder. A second deposition process is required to attach the TEM lamella to the sample holder. Then the lamella is separated from the tip by FIB cut and the lamella is ready for final thinning. As an additional step the W needle has to be sharpened by FIB for the next lamella transfer. This process takes time and could be risky if the welding does not properly work. Additional material from welding has to be removed from lamella, which could be critical, if the region of interest is already reached. Another disadvantage of this method is the fixing of the TEM sample on one side only that can avoid final thinning due to bending which is mostly critical for multi material systems as for GaN HEMT structures.

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To overcome these disadvantages a new fast and reliable lift out process is developed based on clamp holders to easily fix TEM lamellas without the need of a welding process [3]. The clamps consist of two parallel small sized poly Si beams forming a 700 nm slit to carry a pre-thinned lamella. The Si beams are elastically deformed during slip in process of the lamella and the generated adhesion force holds the lamella in the right position. To secure a stable fixing and to avoid overstressing of the beams the lamella has to be thinned to a thickness of 700 nm at the lower side with a vertically slightly increasing thickness of 1–2°. The poly Si clamps are an inherent part of a specific TEM grid geometry which can be directly transferred into a TEM sample holder. The poly Si structure is fabricated by POLYMUMPS processing on a silicon wafer and can be released by wet chemical etching of a sacrificial Si-oxide layer. Because the poly-Si structure is only 3 µm thin it cannot be handled without damaging. To increase the mechanical stability of the structure a specific plating process to grow Cu from both sides is applied and then the whole structure is released from the Si transfer wafer. Several single and double clamp geometries have been designed to be most flexible for different lamella geometries. For GaN multi material system the double slit is used mainly to fix the sample on both sides to increase the mechanical stability and to avoid bending (see Fig. 1).

To transfer the lamella into the clamp holder a Kleindiek lift out shuttle (LOS) equipped with a piezo driven gripper is used. The gripper can be positioned exactly by x, y, z piezo drivers. The open gripper is positioned above the pre-thinned lamella, moved down to the lamella, then closed and lifts the lamella up after FIB cutting. Then the TEM sample is transferred in-situ to the TEM grid and slit into one of the double clamps. The complete transfer process can be done within 5 min only (Fig. 2).

3. Thickness control by using SEM endpoint detection

In combined SEM/FIB systems the preparation progress can be imaged by SEM during FIB milling. If the energy of the primary electrons is high enough to penetrate through the TEM lamella the secondary electron (SE) intensity increases while the amount of back scattered electrons (BSE) decreases. This transparency effect depends on material parameter, primary electron energy, lamella thickness and incident angle of the electron beam which is fixed at 54° given by the tilt angle between SEM and FIB column. The advantage of the observation of the BSE intensity is that it is not affected by FIB generated electrons and therefore can be directly related to the remaining thickness of the lamella (see Fig. 3).

Based on this a generalized and standard free thickness measurement of electron transparent lamellas in the SEM was developed. The method is based on a global normalized transparency

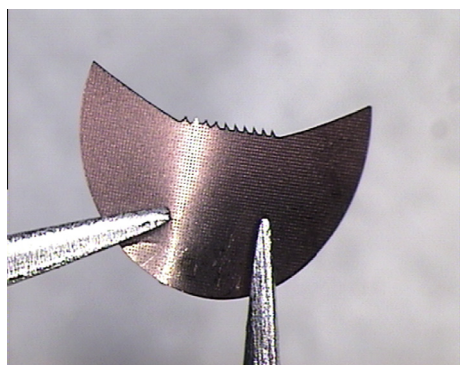


Fig. 1. TEM grid with poly-Si clip holders.

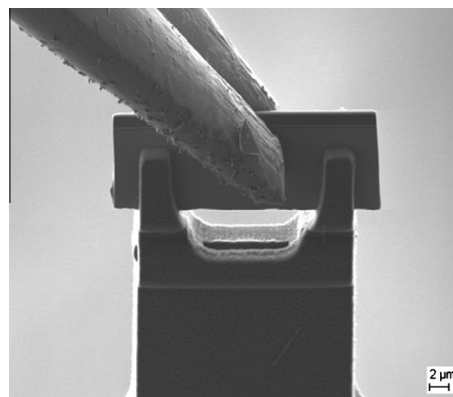


Fig. 2. TEM lamella placed into a double slit holder by using a micro gripper.

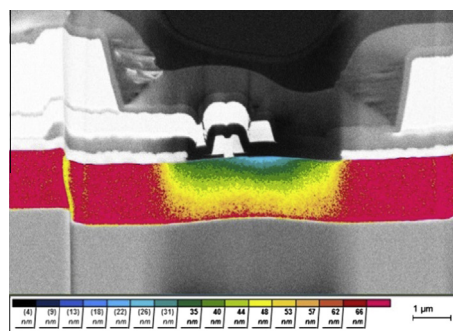


Fig. 3. Example for a BSE thickness map of a HEMT structure taken from the GaN substrate.

function (GNTF) that was derived from Monte-Carlo simulations and tested at wedge shaped samples for typical materials in Si based technologies which are characterized by EELS, CBED and FIB cross sections [2]. The GNTF represents the correlation between the normalized BSE coefficient and the thickness depending on primary electron energy and material parameters. It is used to correlate the lamella thickness with the BSE image intensity with a deviation of the thickness measurement of around 10%.

A software tool was developed to visualize the thickness distribution of a TEM lamella as a color coded map. The software is directly adapted to ZEISS FIB/SEM tools equipped with intense EsB detection. It can be used for mapping the lamella thickness and for end point detection during FIB milling. The remaining thickness of the lamella can be controlled during FIB polishing until the required thickness is reached. This makes the FIB milling process more controllable especially if several samples should have the same thickness to be compared to each other.

4. X2 window preparation of ultra-thin lamellas

Atomic resolution TEM investigations and analytics require samples with residual thicknesses of only several 10 nm. This could be challenging for FIB made lamellas especially for inhomogeneous device structures like HEMTs including different materials, strong topography and possible intrinsic stress. The TEM lamellas typically bend during final thinning because of their low mechanical stability given by the different materials used for HEMTs.

For standard geometries the lamella cannot be mechanically fixed from all sides what could cause strong deformation. A much better situation is given by using the recently developed X2

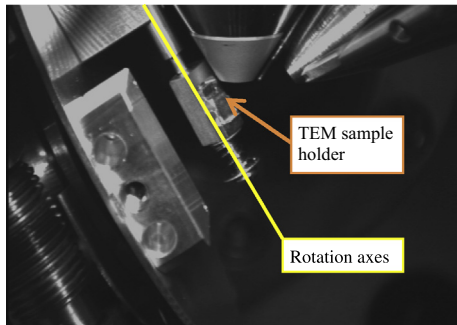


Fig. 4. Sample holder adapted to the Kleindiek RotTipTM to realize the second tilt axis.

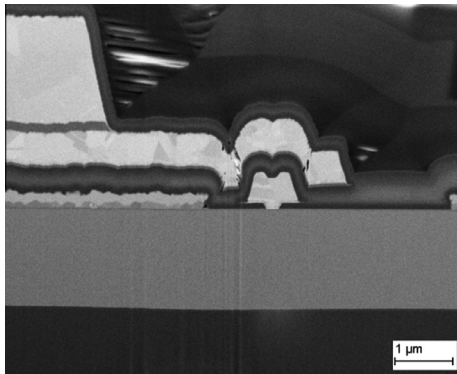


Fig. 5. SEM cross-section of a GaN HEMT device.

window technique [1]. Here the lamella is thinned from both sides under different incident angles producing an electron transparent window which is stabilized by a thicker frame surrounding the thinned window. For this the TEM sample has to be rotated around a second tilt axis perpendicular to the lamella. To realize this a small sized sample holder was prepared and mounted on an additional rotation axis RotTipTM from Kleindiek which is fixing the TEM grids to rotated them some 10° to the right and left, Fig. 4.

5. Optimized FIB/TEM preparation workflow

Fig. 5 shows a typical GaN based HEMT structure. The Au source/drain contacts are several μm high depending on the technology, both are forming a deep cavity, in between the small sized gate structure is positioned. The gate could consist e.g. of Au, Pt and Ni with much lower electron transparency in relation to the SiN-passivation around the gate or the III/V semiconductor substrate underneath.

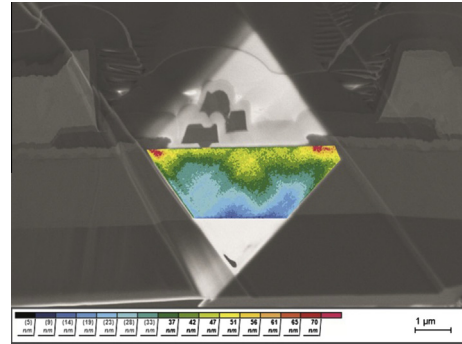


Fig. 7. X2 preparation from top side of the device, GaN/AlGaIn substrate with curtaining from both sides.

The TEM sample preparation techniques are evaluated using fully processed and functional GaN-HEMT devices. As a first step the cavity is filled by FIB deposited carbon to smooth the topography of the structure. Then a > 700 nm thick lamella is made by FIB. If necessary a part of the upper structure is cut off to reduce the lamella topography. After FIB deposition of a protection layer the lamella is transferred by using the Kleindiek LOS micro gripper into the clamp of the TEM grid. Then the TEM grid is mounted to the RotTip stage and the lamella is further thinned by X2 technique with +45° tilt from one and –45° from the other lamella side.

BSE end point detection is used to control the thinning process ending up with low voltage polishing at 5 kV FIB to reduce the sidewall damage reaching a final lamella thickness between 37 nm and 47 nm at the GaN substrate, Fig. 7. There is still some curtaining present at the lamella and therefore, a second sample is made including back side milling of the TEM sample. The same pre-preparation of the lamella was done. Then the lamella is moved by LOS into one slit of a double clamp and next it is tilted and slit top down step by step into the double clamp, Fig. 6.

Now the final thinning could be done from back side of the homogeneous SiC substrate with much less curtaining at the interesting GaN/AlGaIn structures. After X2 milling and final polishing at 5 kV a smoother lamella thickness of only 30–37 nm could be reached, Fig. 8.

6. Examples

6.1. Degradation by “pitting” effect

A typical life time limiting degradation mechanism of HEMT transistor structures is so called “pitting” [4–6]. During operation of a GaN/AlGaIn HEMT transistor a high electrical field is formed within the semiconductor near the drain gate edge. This high electric field causes strain in the AlGaIn barrier due to the inverse piezoelectric effect. This piezoelectric strain induces additional

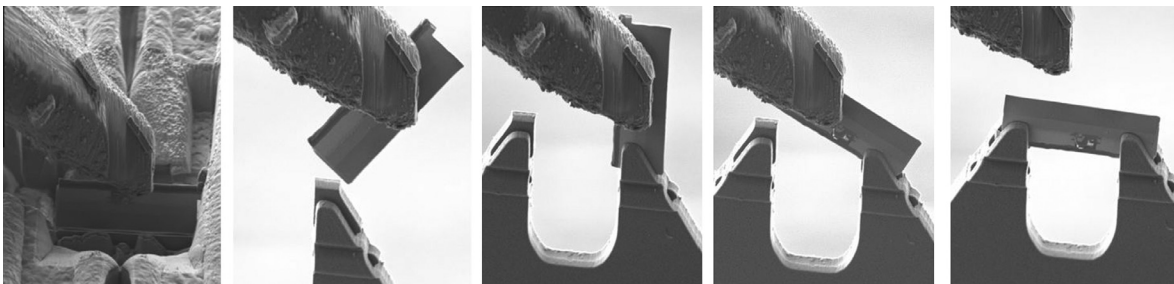


Fig. 6. TEM lamella positioning for backside FIB preparation to reduce curtaining effects.

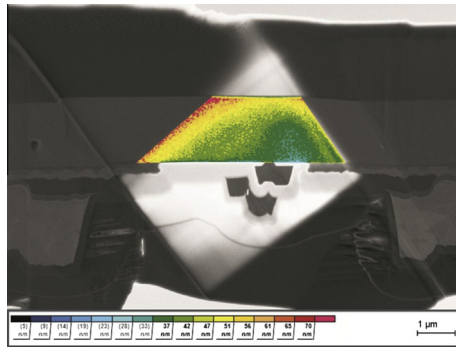


Fig. 8. X2 preparation from back side of the device getting an almost homogeneous thickness in the GaN/AlGaIn substrate.

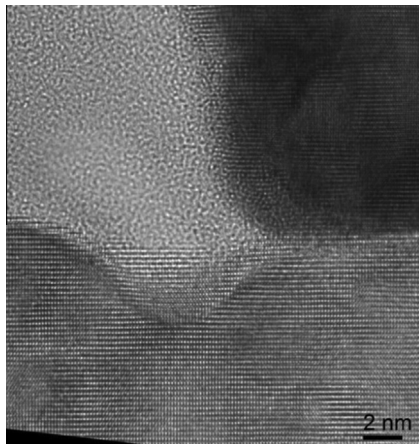


Fig. 9. Atomic resolution TEM image of a pitting defect in AlGaIn barrier at metal gate edge (thickness 35 nm).

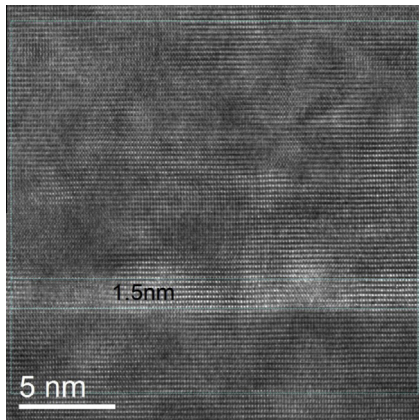


Fig. 10. Method 1: atomic resolution TEM image of AlN spacer layer between GaN and AlGaIn barrier.

mechanical stress between the few nm thin GaN cap layer and AlGaIn barrier on top of the lattice mismatch. This can lead to crystallographic defects near the gate edge due to strain induced diffusion for strain relaxation. These defects are electrically active and behave as additional electron traps. The degradation starts with local pitting and ends with a crack along the gate through the AlGaIn barrier which generates a leakage path between gate and channel.

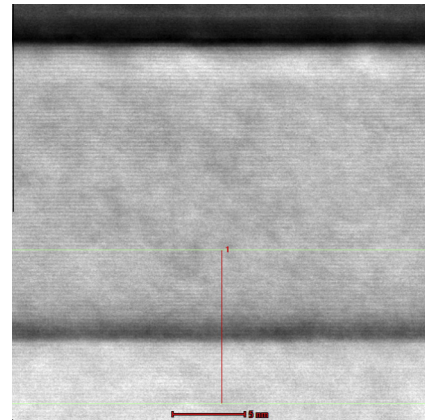


Fig. 11. Method 2: STEM image of GaN–AlN–AlGaIn layer stack with position of line scan.

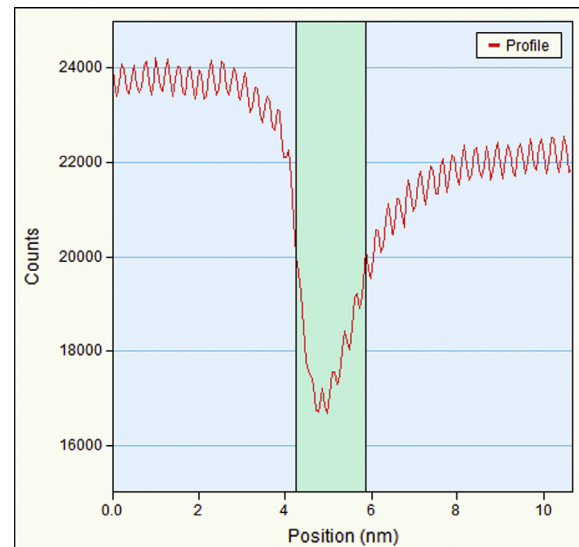


Fig. 12. Method 2: STEM gray value line scan of AlN spacer layer for thickness measurements.

To characterize such crystallographic defects atomic resolution TEM investigation is required because of the small structural lattice defects in the first state of the degradation, Fig. 9. The TEM investigation offers the possibility to characterize the defect formation dependent on the operation related strain or degradation level respectively [4].

6.2. Exact measurement of AlN spacer

To enhance the performance of GaN HEMT transistors a very thin AlN spacer can be introduced between GaN buffer and AlGaIn barrier layer. The AlN increases the two-dimensional electron gas (2DEG) density, mobility and drain current because it reduces the strain of the AlGaIn layer with enhanced film quality. Only a few atom layers of AlN are needed to enhance the performance of HEMT devices. If AlN layer is too thick performance drops down due to its insulation behaviour [7,8]. Therefore, it is important to control the spacer layer thickness and composition. High resolution TEM imaging and analytics with high spatial resolution is used to measure the AlN layer thickness with sub nm precision. This is challenging because of the weak contrast between GaN, AlN and

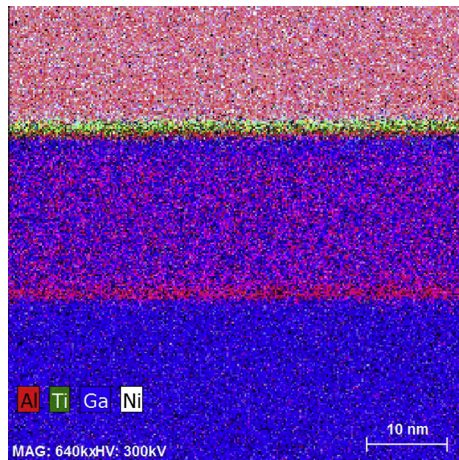


Fig. 13. Method 3: STEM–EDX mapping of GaN–AlN–AlGaIn–GaN–metal layer stack with element distribution.

AlGaIn due to the same lattice structures. For that reason three different TEM imaging techniques were evaluated and compared to get best results for AlN thickness measurement by using FEI Titan G3 high resolution TEM with image correction and Super-X EDX detector. HRTEM images, Fig. 10, show a brighter contrast for the AlN due to the lower mass thickness of the AlN compared to GaN and AlGaIn. The phase contrast of the slightly disturbed FIB cross section causes some contrast variations in the lattice image. High resolution STEM imaging, Fig. 11, is less sensitive for lattice distortions and the dark contrast of the AlN between the surrounding Ga containing nitrides can be seen more easily. The interface between the lower GaN and the AlN is sharper than the interface between the AlN and the top AlGaIn. Using half of the contrast change as criterion to localize the AlN/AlGaIn interface the measured thickness of the AlN layer is 1.6 nm with an uncertainty of 0.26 nm (1/2 lattice distance in *c*-direction), Fig. 12. Element mappings with the high sensitive Super-X EDX detector illustrate the sharpness of the GaN/AlN interface, Fig. 13. Due to the low signal to noise ratio

the accuracy of the thickness measurements in EDX mappings is worse than measurements in STEM images.

7. Conclusion

In this paper it is shown how a very thin and homogeneous TEM sample can be prepared by using a crossbeam FIB station in combination with a Kleindiek lift-out shuttle and SEM thickness end point detection. To reach an almost homogeneous extremely thin and artefact free lamella for high resolution TEM analysis the lamella is placed upside down in a specific designed clamp holder and finally thinned from backside by X2 method down to a few 10 nm thickness.

The feasibility of this preparation flow is demonstrated in two application examples for GaN HEMT device atomic resolution characterization of pitting failure mode and precise measurement of the AlN spacer thickness with sub nm resolution.

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